

MANISH PATODIYA

AI Engineer | GenAI & Agentic AI Systems

Bengaluru, India | +91 96020 65385 | mpatodiya2@gmail.com | linkedin.com/in/manish-patodiya | Portfolio

SUMMARY

AI Engineer with experience building enterprise **LLM applications**, **AI agents**, **RAG systems**, and **agentic workflows**. Skilled in **LangGraph**, **LangChain**, **hybrid retrieval**, **prompt engineering**, **LLM evaluation**, and **AWS Bedrock**. Experienced delivering **production-grade AI solutions** for enterprise automation.

CORE SKILLS

AI/GenAI: LLMs, Agentic AI, AI Agents, LangGraph, LangChain, RAG, GraphRAG, Hybrid Retrieval, Tool Calling, Prompt Engineering, BM25, Vector Search, Knowledge Graphs, Vector Databases (pgvector, pinecone, chroma, SAP HANA).

Languages & Backend: Python, SQL, FastAPI, PostgreSQL, SAP HANA, Docker, Node.js, Flask

Cloud & Tools: AWS Bedrock, SAP AI Core, SAP BTP, Git, Playwright, Cursor, Claude Code, GitHub Copilot

Frontend: React, Next.js, TypeScript

EXPERIENCE

AI Engineer – Research & Engineering, Banking

Mar 2024 – Present

MSG Global Solutions

Bengaluru, India

- Leading development of an enterprise AI platform for **BAIS** featuring an **LLM-powered Knowledge Assistant** and **SmartFilter** for automated regulatory reporting validation.
- Developed **Snow-Vision**, an **enterprise GenAI platform** for SAP PAPM enabling intelligent incident analysis and workflow automation.
- Built an automated **LLM evaluation framework** to benchmark retrieval quality, hallucination rates, answer correctness, and agent execution across enterprise datasets.
- Built production AI applications using **AWS Bedrock** and **SAP AI Core** with scalable inference and prompt orchestration.

Full Stack Developer

Oct 2021 – Oct 2022

GeeksIT Data Solutions

Jaipur, India

- Led a team to build 3 full-stack applications, developed 100+ REST APIs improving performance by 30–40%, and resolved 120+ production issues, significantly enhancing platform stability and user experience.

PROJECTS

AI Regulatory Intelligence Platform for MSG BAIS - Python, LangGraph, GenAI, AI Agents, Next.js, PostgreSQL, FastAPI

Module 1 — Knowledge Assistant (Enterprise RAG)

- Built an **enterprise knowledge platform** enabling natural language reasoning over BAIS documentation, PostgreSQL schema, validation rules, and regulatory reference data.
- Designed **hybrid retrieval** using **LangGraph**, **pgvector**, metadata filtering, and **BM25** to deliver **citation-backed regulatory responses**.

Module 2 — SmartFilter (Rule + AI Correction Engine)

- Developed **SmartFilter**, an **AI-driven validation correction engine** that analyzes BAIS validation failures and recommends attribute-level corrections before regulatory submission.
- Designed correction pipelines using **deterministic validation logic**, master-data lookups, fuzzy matching, **web search**, and **GLEIF enrichment**.
- Architected scalable validation processing by grouping failures by validation type instead of **per-record LLM reasoning**.

Snow-Vision — Enterprise GenAI Automation Platform - GenAI, Python, LangGraph, AI Agents, Next.js, Node.js, SAP S/4HANA

- Architected an **enterprise Generative AI platform** that unified enterprise knowledge, incident history, and operational workflows to support intelligent reasoning and automated actions across SAP PAPM.
- Built **LangGraph agent workflows** with **tool calling** to orchestrate retrieval, web search, database queries, and enterprise APIs.
- Designed scalable **SAP HANA-based memory architecture** with automated ingestion (**50+ tickets/day**), alerts, dashboards, and analytics.
- Developed **LLM evaluation frameworks** and deployed production AI workloads on **SAP AI Core** and **SAP BTP**.

EDUCATION

Master of Computer Applications (MCA) | CGPA: 9.49

2022 – 2024